ML Builder Expo

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

ML Builder Expo

> Prepare ML Builder Expo: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Defend the project using traceable **evidence**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Capstone Train, Analyze, Improve

Without opening the old lesson, explain one key idea from Capstone Train, Analyze, Improve. Give an example and a mistake that the idea helps you avoid.

Look for: Run a frozen final evaluation with complete accounting

Reference example, not the only acceptable wording: Teacher explanation: Candidate accuracy 9/12=75 percent, precision 2/3, recall 2/4; baseline 8/12 about 66.7 percent, with all four blocked cases missed.

Model Cards and Human Review

Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.

Look for: Write a model card whose claims match **evidence**

Reference example, not the only acceptable wording: Teacher explanation: Intended: compare toy classifiers on the documented fictional task.

Not supported: real parcel handling, student assessment or decisions outside the evaluated setup.

Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Final counts are TP=12,FN=3,FP=5,TN=20. Compute accuracy, precision and recall, then state one **limitation** and a review trigger for the model **defense**.

Concept to explain

A builder defends the complete **evidence** chain: problem, data, code, model, metrics, errors, fairness, **limitation**s, and human oversight.

Term	Meaning
technical defense	evidence-based explanation under questions
traceability	link claim to data, code, result, and decision
transfer test	fresh case not rehearsed
release decision	approve, limit, revise, or stop

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and **reflection** 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on machine learning project **defense**. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

A result describes observed **evidence**; a claim must not extend beyond what that **evidence** supports.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

End goal for the pictorial case

By the end, I can explain ml builder expo using a traced example and a stated **limitation**.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model ML Builder Expo using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Recompute a fresh metric and answer an unseen failure case during the **defense**.

MODEL IDEA A builder defends the complete **evidence** chain: problem, data, code, model, metrics, errors, fairness, **limitation**s, and human oversight.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated **limitation**.

How the mechanism works

UNDER THE HOOD Trace one leaf from source and split through pre**process**ing, model, **prediction**, and review. Recompute precision or recall from a fresh confusion matrix without notes.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Trace one leaf from source and split through pre**process**ing, model, **prediction**, and review.

Step 2

STEP 2 Recompute precision or recall from a fresh confusion matrix without notes.

Step 3

STEP 3 Run an unseen low-light or missing-feature case and follow the documented fallback.

Step 4

STEP 4 State intended use, do-not-use, subgroup **limitation**, provenance, and who remains accountable.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

Final classifier gets16/20 correct; two of four errors are false negatives.

The teaching case

Report supported metrics and what is missing.

Exact explanation to model

Why the result follows

Accuracy**80%**. Recall cannot be computed from FN alone; actual positives or TP are needed. Show full matrix, limits and review path.

Explain the added worked example

Explain the added worked example · 1

The final **defense** tells a coherent story: question, data, method, comparison, errors and limits. A strong project may conclude that the baseline is preferable or **evidence** is insufficient.

The aim is sound reasoning and a reproducible investigation, not an impressive headline or polished unsupported claim.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines machine learning project **defense**.

Explain the added worked example · 2

Example opening: We compared a simple classifier with an always-clear **training** baseline on fictional parcels. We froze the procedure before a 12-case test. Candidate accuracy was 9/12 versus baseline 8/12; candidate FP 1 FN 2.

These small synthetic results support a classroom comparison, not real parcel decisions. We would next investigate dim-condition coverage using new development data.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Key terms

Term	Meaning
Defense	An explanation of decisions supported by inspectable evidence
Traceability	Ability to follow a claim back to its source
Limitation	A boundary on interpretation or supported use
Reflection	An account of learning and justified next steps

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Machine learning project defense at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Show the process	Task data splits versions	Explain one prediction
Show the evidence	Baseline metrics errors	Trace one reported number
Show the limits	Unsupported uses next step	Answer peer questions

Prepare a five-minute presentation with a one-page evidence summary and linked project files. Include the data dictionary, split record, run record, prediction ledger, model card and revision history. Demonstrate one prediction or arithmetic calculation live so the audience can follow the reasoning.

Name the files needed to audit one reported accuracy figure. Explain the steps or evidence behind your response.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: ML Builder Expo

Present a plant-classifier expo claim that another team could check.

The facts we will use

> **Supplied facts**
>
> Toy expo **evidence**: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct. Three candidate failures are recorded; no outdoor trial exists.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/36

ML Builder Expo

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Show denominator

Candidate12/15=80%; baseline10/15≈66.7% on the same cases.



2. Show independence

Explain five held-out plant identities and three related views per plant.



3. Show failures

Present the three failure records and indoor-only limitation beside the headline rate.

CHANGE ONE THING

The poster omits the baseline and calls the model "field ready".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Show denominator	Candidate12/15=80%; baseline10/15≈66.7% on the same cases.
2. Show independence	Explain five held-out plant identities and three related views per plant.
3. Show failures	Present the three failure records and indoor-only limitation beside the headline rate.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Clear presentation helps review but does not increase the strength of a tiny or unrepresentative evaluation.

Change one thing in the pictured case

Change one thing

Change: The poster omits the baseline and calls the model “field ready”.

Predicted result

Prediction: The headline exceeds the supplied **evidence**.

Why the result follows

Reason: No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Ask while showing the picture

Ask while showing the picture · 3

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

Demonstrate ML Builder Expo, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Board sequence

1. Put the facts on the board

Write the given facts: Final classifier gets16/20 correct; two of four errors are false negatives.

2. State the decision to explain

Write the question to explain: Report supported metrics and what is missing.

3. Reveal the worked reasoning

Reveal after **prediction**s: Accuracy**80%**. Recall cannot be computed from FN alone; actual positives or TP are needed. Show full matrix, limits and review path.

Hold back the independent task

Keep for an independent attempt: Your expo model succeeds on three demonstration plants.

What **evidence** should accompany the demo and what **limitation** should you state?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: **Prediction** ledger with references and **output**s, split membership, frozen model/run record and metric definition; count correct rows over all scored rows under the saved policy.

Exit response: A valid response links a documented **limitation** to new development **evidence**, records any revised version, and reserves fresh independent final-test **evidence** for a later frozen evaluation.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: What is the difference between a result and a claim? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Review the four rubric dimensions. Use W1–W3 to prepare a bounded opening and an auditable evidence summary.

Stage: 18 to 32 minutes

Teacher action: Teams prepare a five-minute defense and one-page summary. Check every claim against the saved project files and rehearse one prediction trace.

Stage: 32 to 45 minutes

Teacher action: Run peer rehearsals in pairs. Each reviewer records a question about data roles, a failure and the claim limits.

Stage: Session 2 start to 8 minutes

Teacher action: Complete W4 independently as a short critique of unsupported claims. Use the result to target final feedback.

Stage: 8 to 20 minutes

Teacher action: Begin defenses in parallel small groups: five minutes presenting plus two minutes of questions per team. Teacher samples evidence and collects all records.

Stage: 20 to 35 minutes

Teacher action: Continue defenses and record scores separately for data/splits, reproducibility, evaluation and communication. Use written evidence as well as the live explanation.

Stage: 35 to 45 minutes

> **Teacher action:** Complete W5 as an individual reflection. Resolve or record peer feedback, identify a justified next step and use the fresh retry for learners who confuse model improvement with project success.

Reduce help deliberately

Completion practice

Completion task: Name the files needed to audit one reported accuracy figure.

Reasoning to listen for

Teacher check: Teacher explanation: **Prediction** ledger with references and **output**s, split membership, frozen model/run record and metric definition; count correct rows over all scored rows under the saved policy.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Machine learning project **defense**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record.

> For this week: A valid response names the fictional task, reports an auditable baseline/candidate comparison with denominator, and restricts the claim to the evaluated setup.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

Show split/provenance, candidate-versus-baseline counts, and actual error records with the indoor-only scope. Changed-case check: The headline exceeds the supplied **evidence**.

No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Changed-case reasoning: The headline exceeds the supplied **evidence**. No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the ML Builder Expo investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Expo **Defense**: present the **evidence** bundle, accept a fresh calculation and transfer case, explain one failure, and defend a **release decision**.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome

Record	What to capture
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR A smooth prepared demo proves the model is reliable and ready for use.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record.

For this week: A valid response names the fictional task, reports an auditable baseline/candidate comparison with denominator, and restricts the claim to the evaluated setup.

A stronger claim makes a stronger project: A valid response links a documented **limitation** to new development **evidence**, records any revised version, and reserves fresh independent final-test **evidence** for a later frozen evaluation.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Request the baseline on the same cases, split/development history and full error counts/limits.

A polished presentation cannot establish validity or supported use without auditable **evidence**.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the ML Builder Expo questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Accuracy**80%**. Recall cannot be computed from FN alone; actual positives or TP are needed. Show full matrix, limits and review path.

Independent question

Independent question · withhold the key

Your expo model succeeds on three demonstration plants. What **evidence** should accompany the demo and what **limitation** should you state?

Expected independent answer

Teacher key · Expected independent answer

Present dataset and split records, baseline, held-out results, errors, subgroup checks, revisions and model-card limits. Three demonstrations do not establish performance across all plants.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A smooth demonstration replaces an independent evaluation and model card.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Total=40; accuracy=32/40=**80%**; precision=12/17≈70.6%; recall=12/15=**80%**. These describe the tested set only. Show conditions and subgroup counts, and route missing or out-of-scope **input**s to review under a predefined policy.

Explanations for the additional workbook examples

W1 Write an opening

Workbook question · W1

Draft a three-sentence opening with purpose, a result and a limit.

Teacher reasoning · W1

Teacher explanation: A valid response names the fictional task, reports an auditable baseline/candidate comparison with denominator, and restricts the claim to the evaluated setup.

W2 Trace evidence

Workbook question · W2

Name the files needed to audit one reported accuracy figure.

Teacher reasoning · W2

Teacher explanation: **Prediction** ledger with references and **output**s, split membership, frozen model/run record and metric definition; count correct rows over all scored rows under the saved policy.

W3 Answer critique

Workbook question · W3

A peer asks whether your model works for every school. Give a supported response.

Teacher reasoning · W3

Teacher explanation: No universal conclusion is supported. State the actual synthetic dataset and conditions, missing coverage and what additional appropriate **evidence** would be needed.

W4 Apply independently

Workbook question · W4

Review a **defense** claiming 95 percent success on 20 cases but omitting the baseline, split history and errors. Give three revision requests and explain why polish is insufficient.

Teacher reasoning · W4

Teacher explanation: Request the baseline on the same cases, split/development history and full error counts/limits. A polished presentation cannot establish validity or supported use without auditable **evidence**.

W5 Plan the next step

Workbook question · W5

Write one next investigation tied to an observed **limitation**, and identify what must stay held out.

Teacher reasoning · W5

Teacher explanation: A valid response links a documented **limitation** to new development **evidence**, records any revised version, and reserves fresh independent final-test **evidence** for a later frozen evaluation.

Assess independent transfer

Independent question · withhold the key

Selected case: Review a **defense** claiming 95 percent success on 20 cases but omitting the baseline, split history and errors. Give three revision requests and explain why polish is insufficient.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Request the baseline on the same cases, split/development history and full error counts/limits. A polished presentation cannot establish validity or supported use without auditable **evidence**.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect ML Builder Expo to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Deliver the final Responsible Plant Classifier product card, live **evidence** trace, model card, and human-oversight commitment.

Keep the decision, **input**, result, explanation and remaining **limitation** in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Review the full project **evidence** and choose concepts needing another independent attempt before the next level. One expo is not proof of mastery of all AI.

Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **defense** and **traceability** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Request the baseline on the same cases, split/development history and full error counts/limits. A polished presentation cannot establish validity or supported use without auditable **evidence**.

Fresh retry: a learner says our baseline won, so the project failed. Ask for a **defense** explaining the valid comparison, retained baseline, limits and an **evidence**-based next investigation; do not reward invented improvement.

Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Level 4 (Grade 9) is complete. Use the project **defense** and independent work to assess readiness for the next level. Revisit split integrity, model **fitting** and evaluation before advancing if those foundations are not yet secure.

Plan the delayed return

Suggested return points in this level: plan a review after the level ends. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed ML Builder Expo lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Defend the project using traceable **evidence**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Name the files needed to audit one reported accuracy figure.

Supported practice answer

Reasoned practice answer: Teacher explanation: **Prediction** ledger with references and **output**s, split membership, frozen model/run record and metric definition; count correct rows over all scored rows under the saved policy.

Check every learner before discussion: Use the worked case for Machine learning project **defense**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record.

For this week: A valid response names the fictional task, reports an auditable baseline/candidate comparison with denominator, and restricts the claim to the evaluated setup.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The poster omits the baseline and calls the model “field ready”. Require a **prediction** and a reason before comparing. Expected result: The headline exceeds the supplied **evidence**.

Reason: No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner’s explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

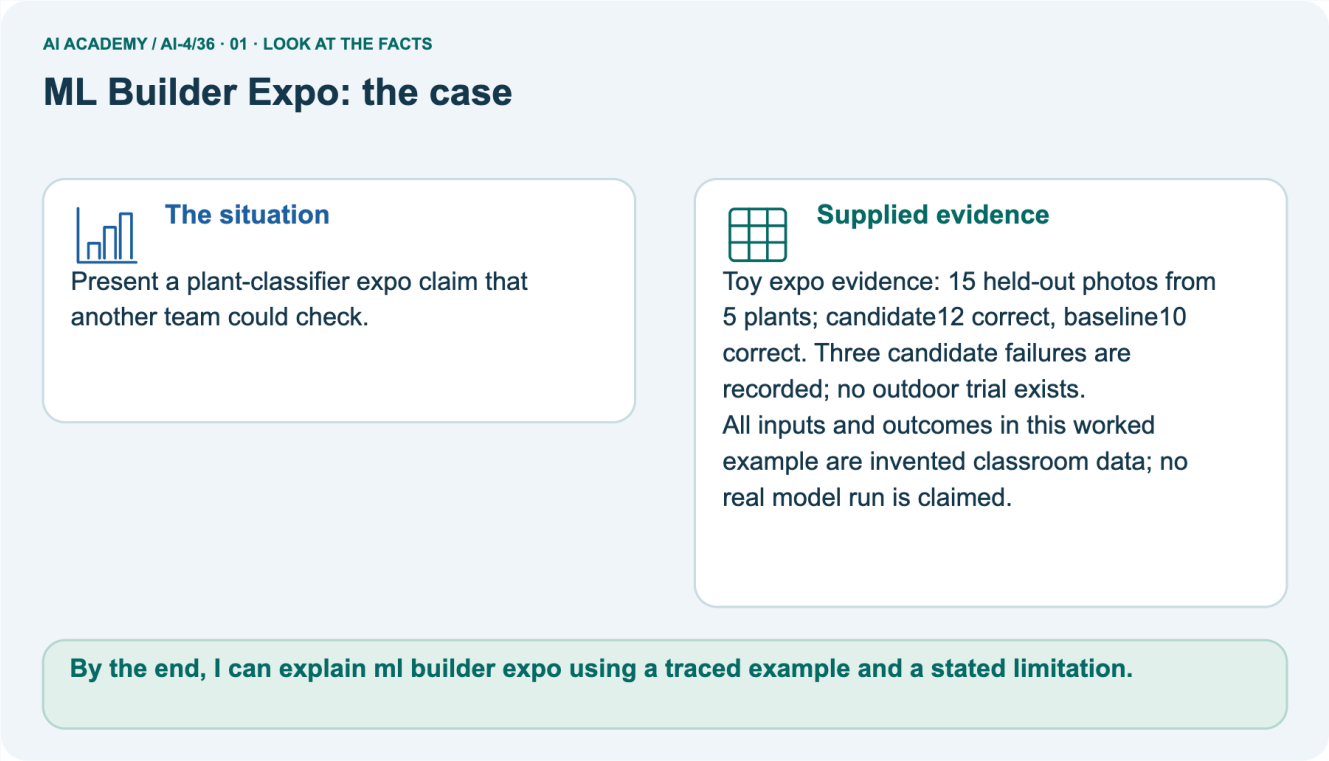
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and **limitation**s. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one ML Builder Expo graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. ML Builder Expo: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Present a plant-classifier expo claim that another team could check. Supplied facts: Toy expo **evidence**: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct.

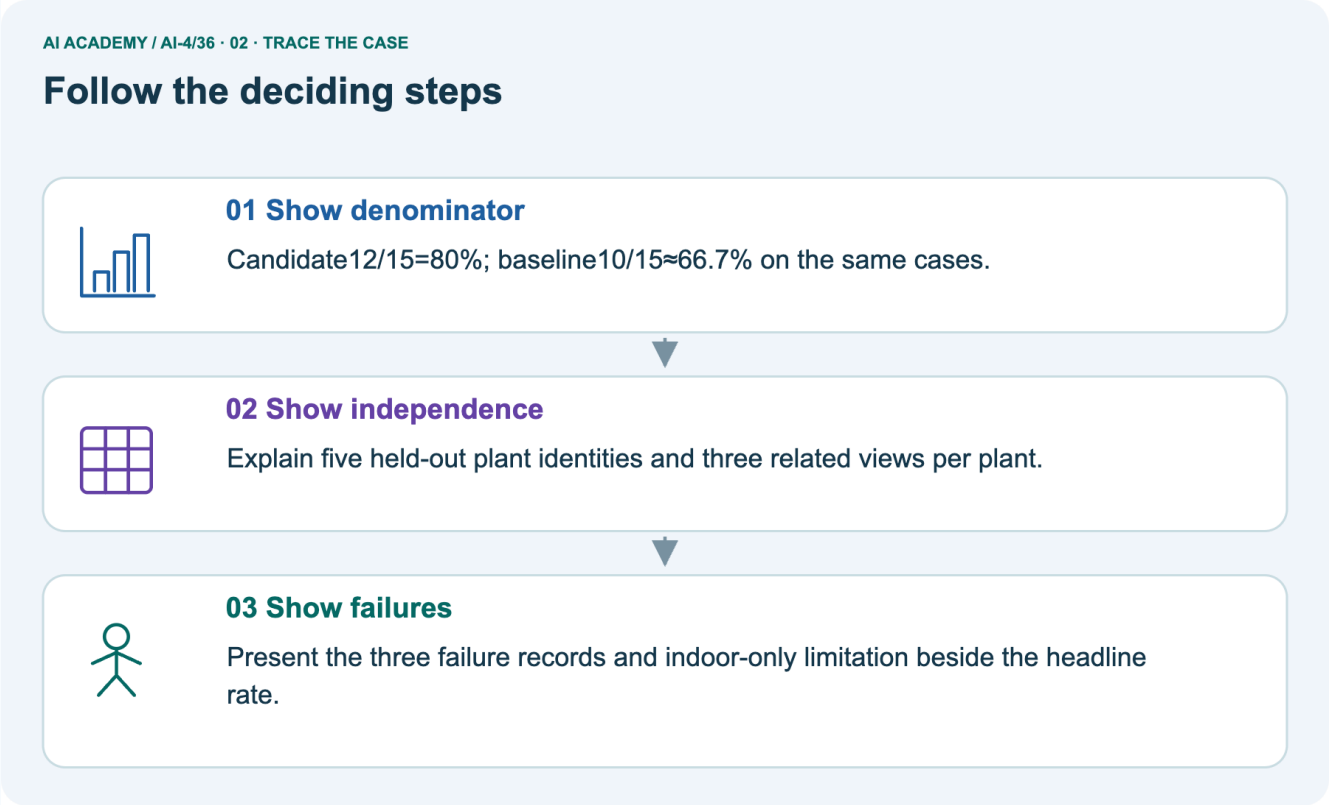
Three candidate failures are recorded; no outdoor trial exists. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

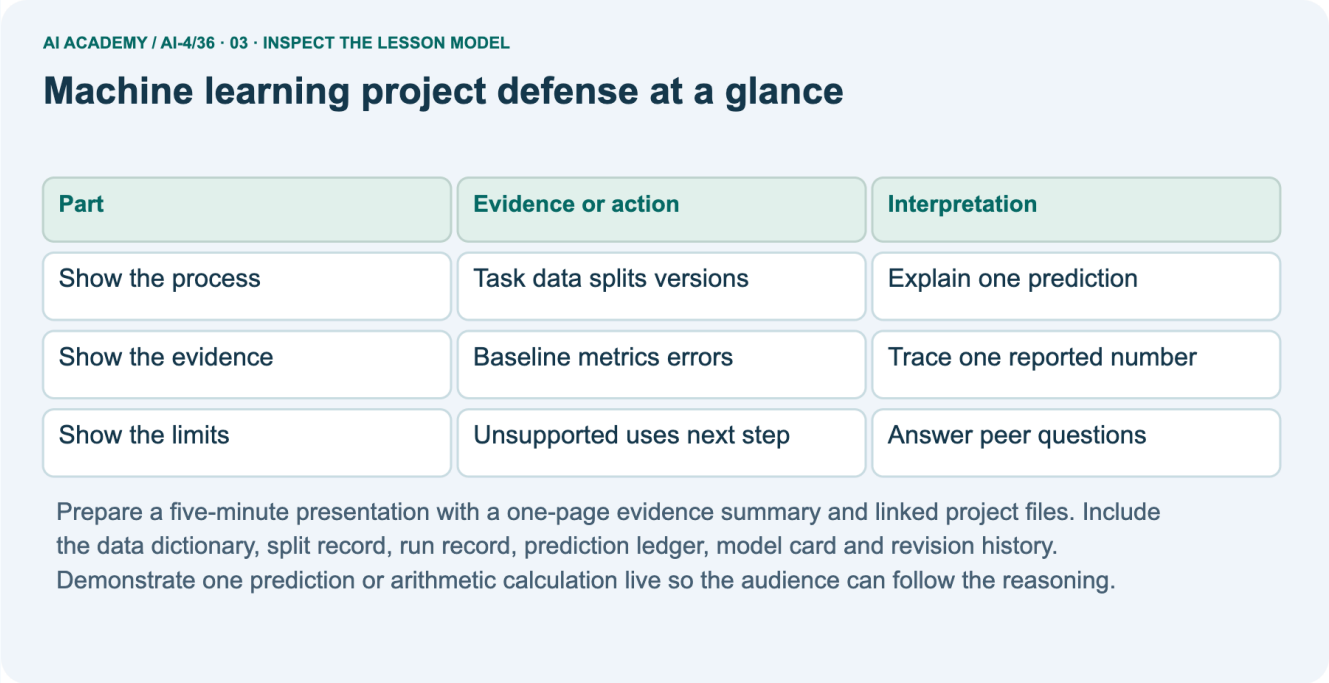
Show denominator: Candidate12/15=**80%**; baseline10/15≈66.7% on the same cases. → Show independence: Explain five held-out plant identities and three related views per plant. → Show failures: Present the three failure records and indoor-only **limitation** beside the headline rate.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Machine learning project defense at a glance



Follow the labelled picture with the matching explanation. Open the vector diagram

Machine learning project **defense** at a glance. Show the **process** | Task data splits versions | Explain one **prediction**. Show the **evidence** | Baseline metrics errors | Trace one reported number.

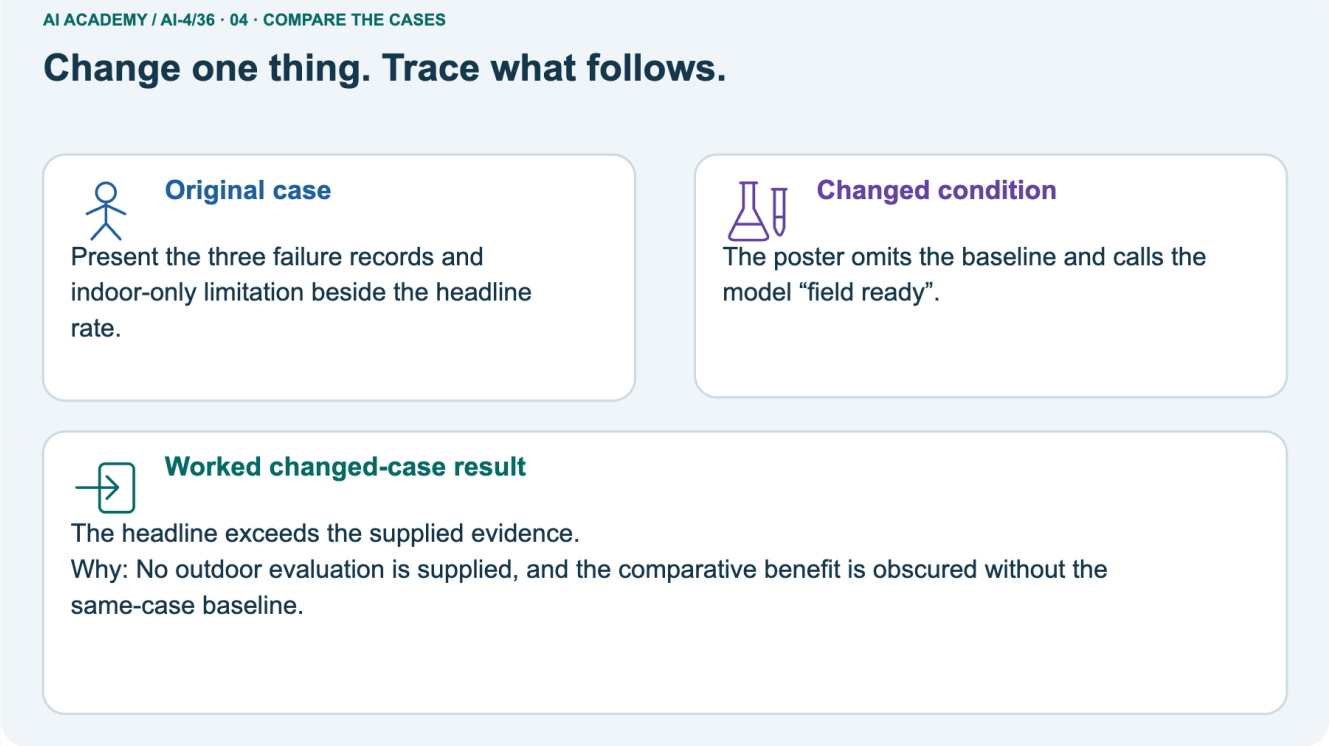
Show the limits | Unsupported uses next step | Answer peer questions

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Present the three failure records and indoor-only **limitation** beside the headline rate. Changed condition: The poster omits the baseline and calls the model “field ready”.

Expected result: The headline exceeds the supplied **evidence**. Reason: No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Predict before checking the worked changed-case result. Name what changed and what the

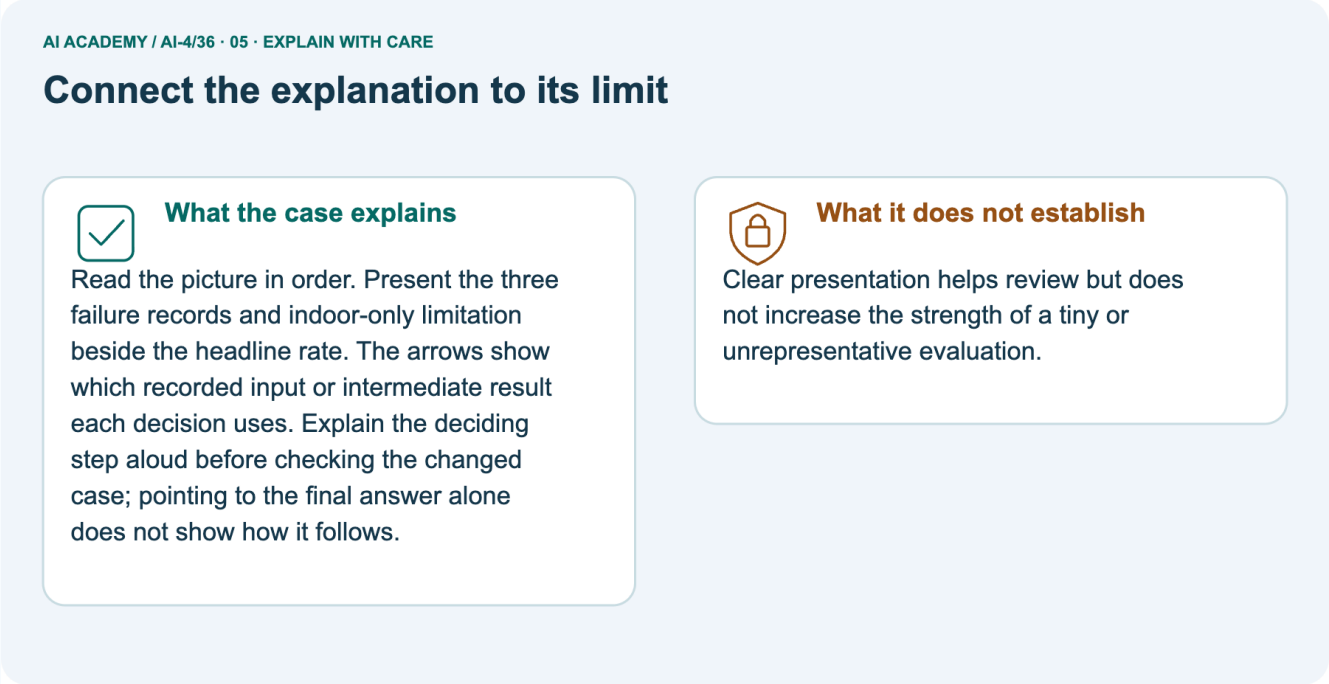
supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Clear presentation helps review but does not increase the strength of a tiny or unrepresentative evaluation.

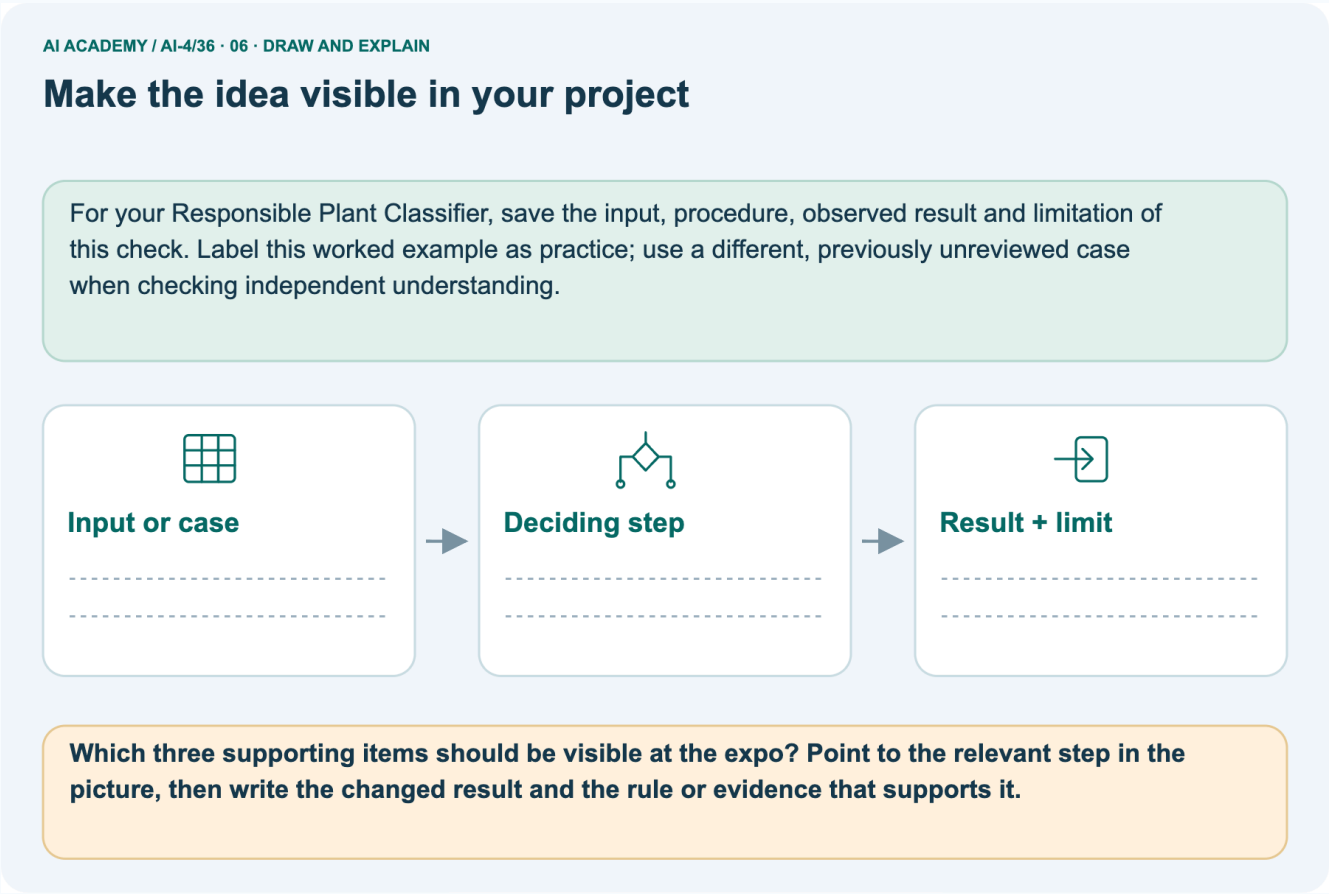
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Clear presentation helps review but does not increase the strength of a tiny or unrepresentative evaluation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Show split/provenance, candidate-versus-baseline counts, and actual error records with the indoor-only scope. Changed-case check: The headline exceeds the supplied **evidence**.

No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.